

Continuous Assessment 3

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Question 1

Let $X \sim N_p(\mu, \Sigma)$ with $|\Sigma| > 0$.

RTP:

$$(X - \mu)' \Sigma^{-1} (X - \mu) \sim \chi_p^2$$

Using the eigen decomposition of Σ :

$$\Sigma = U \Lambda U'$$

where U is an orthogonal matrix, such that

$$U' = U^{-1}$$

Then

$$\Sigma^{-1} = (U \Lambda U')^{-1} = (U')^{-1} \Lambda^{-1} U^{-1} = U \Lambda^{-1} U'$$

Substituting in Σ^{-1}

$$\begin{aligned} (X - \mu)' \Sigma^{-1} (X - \mu) &= (X - \mu)' U \Lambda^{-1} U' (X - \mu) \\ &= (X - \mu)' U \Lambda^{-\frac{1}{2}} \Lambda^{-\frac{1}{2}} U' (X - \mu) \end{aligned}$$

Let

$$Y = \Lambda^{-\frac{1}{2}} U' (X - \mu)$$

and let

$$A = \Lambda^{-\frac{1}{2}} U'$$

Therefore

$$Y = A(X - \mu) \sim N_p(0, A\Sigma A')$$

Solve for the Covariance matrix of Y

$$\begin{aligned} A\Sigma A' &= (\Lambda^{-\frac{1}{2}}U') \Sigma (\Lambda^{-\frac{1}{2}}U')' \\ &= (\Lambda^{-\frac{1}{2}}U') \Sigma U(\Lambda^{-\frac{1}{2}})' \\ &= \Lambda^{-\frac{1}{2}}U' U \Lambda U' U(\Lambda^{-\frac{1}{2}})' \\ &= \Lambda^{-\frac{1}{2}}\Lambda (\Lambda^{-\frac{1}{2}})' \\ &= I_p \end{aligned}$$

Y has a Multivariate Standard Normal Distribution.

$$Y \sim N_p(0, I_p)$$

This means that individual Y_i components have a Standard Normal Distribution.

$$Y_i \sim N(0, 1)$$

The Sum of Squares of independent Standard Normal variables has a Chi-squared Distribution with p degrees of freedom.

$$Y'Y = \sum_{i=1}^p Y_i^2$$

$$\sum_{i=1}^p Y_i^2 \sim \chi_p^2$$

Since

$$(X - \mu)' \Sigma^{-1} (X - \mu) = Y'Y$$

We have that

$$(X - \mu)' \Sigma^{-1} (X - \mu) \sim \chi_p^2$$

Question 2

We are given μ defined as

$$\mu = \begin{bmatrix} 5 \\ 0 \\ -2 \\ 6 \\ 2 \end{bmatrix}$$

Rearranging the rows to how X1 and X2 are defined, results in two μ vectors

$$\mu_1 = \begin{bmatrix} 5 \\ 0 \\ 6 \end{bmatrix} \quad \mu_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 8 & 3 & 0 & -1 & 5 \\ 3 & 12 & 2 & 2 & -2 \\ 0 & 2 & 8 & 0 & 2 \\ -1 & 2 & 0 & 9 & 1 \\ 5 & -2 & 2 & 1 & 10 \end{bmatrix}$$

Σ is partitioned such that,

$$\Sigma_{11} = \begin{bmatrix} 8 & 3 & 0 \\ 3 & 12 & 2 \\ 0 & 2 & 8 \end{bmatrix} \quad \Sigma_{12} = \begin{bmatrix} -1 & 5 \\ 2 & -2 \\ 0 & 2 \end{bmatrix} \quad \Sigma_{21} = \begin{bmatrix} -1 & 2 & 0 \\ 5 & -2 & 2 \end{bmatrix} \quad \Sigma_{22} = \begin{bmatrix} 9 & 1 \\ 1 & 10 \end{bmatrix}$$

To calculate the value of $\mu_{1|2}$ we use

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

From this we now just need to calculate the values of $(x_2 - \mu_2)$ and Σ_{22}^{-1} :

$$x_2 - \mu_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The inverse of Σ_{22} is

$$\Sigma_{22}^{-1} = \begin{bmatrix} \frac{10}{89} & -\frac{1}{89} \\ -\frac{1}{89} & \frac{89}{89} \end{bmatrix}$$

Now that we have values for all the parameters, $\mu_{1|2}$ can easily be computed to give us:

$$\mu_{1|2} = \begin{bmatrix} \frac{430}{89} \\ \frac{22}{89} \\ \frac{532}{89} \end{bmatrix}$$

To get the covariance matrix we use the formula: $\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$

Resulting in a 3x3 matrix:

$$\Sigma_{1|2} = \begin{bmatrix} \frac{467}{89} & \frac{389}{984} & -\frac{92}{218} \\ \frac{389}{984} & \frac{89}{218} & \frac{89}{676} \\ -\frac{92}{218} & \frac{89}{676} & \frac{89}{89} \end{bmatrix}$$

We now have all the elements for the conditional (joint) distribution

$$X_1 | X_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \sim N_3(\mu_{1|2}, \Sigma_{1|2})$$

The conditional (joint) distribution now can be written as,

$$X_1 | X_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \sim N_3\left(\begin{bmatrix} \frac{430}{89} \\ \frac{22}{89} \\ \frac{532}{89} \end{bmatrix}, \begin{bmatrix} \frac{467}{89} & \frac{389}{984} & -\frac{92}{218} \\ \frac{389}{984} & \frac{89}{218} & \frac{89}{676} \\ -\frac{92}{218} & \frac{89}{676} & \frac{89}{89} \end{bmatrix}\right)$$

Simplified:

$$X_1 | X_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \sim N_3\left(\begin{bmatrix} 4.831 \\ 0.247 \\ 5.978 \end{bmatrix}, \begin{bmatrix} 5.247 & 4.371 & -1.034 \\ 4.371 & 11.056 & 2.449 \\ -1.034 & 2.449 & 7.596 \end{bmatrix}\right)$$